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Key Points:

- High variability in spring nitrate hysteresis-shape contrasts single-shape dominance in surface streams
- Rapid versus slow connection of nitrate to the karst spring is primarily a function of antecedent aquifer conditions
- Integrating high-frequency sensing into numerical model evaluation reduces prediction uncertainty

Supporting Information:

Supporting Information may be found in the online version of this article.

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Nitrate Hysteresis as a Tool for Revealing Storm-Event Dynamics and Improving Water Quality Model Performance

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Abstract Understanding the physics of nitrate contamination in surface and subsurface water is vital for mitigating downstream water quality impairment. Though high frequency sensor data have become readily available and computational models more accessible, the integration of these two methods for improved prediction is underdeveloped. The objective of this study was to utilize high-frequency data to advance our understanding and model representation of nitrate transport for an agricultural karst spring in Kentucky, USA. We collected 2-years of 15-min nitrate and specific conductance data and analyzed source-timing dynamics across dozens of events to develop a conceptual model for nitrate hysteresis in karst. Thereafter, we used the sensing data, specifically discharge-concentration indices, to constrain modeled nitrate prediction bounds as well as the uncertainty of hydrologic and nitrogen processes, such as soil percolation and biogeochemical transformation. Observed nitrate hysteresis behavior at the spring was complex and included clockwise ($n = 11$), counterclockwise ($n = 13$), and figure-eight ($n = 10$) shapes, which contrasts with surface systems that are often dominated by a single hysteresis shape. Sensing results highlight the importance of antecedent connectivity to nitrate-rich storages in determining the timing of nitrate delivery to the spring. After integrating hysteresis analysis into our numerical model evaluation, simulated nitrate prediction bounds were reduced by 43 ± 12% and parameter uncertainty by 36 ± 20%. Taken together, this study suggests that discharge-concentration indices derived from high-frequency sensor data can be successfully integrated into numerical models to improve process representation and reduce modeled uncertainty.

Plain Language Summary Understanding how contamination arrives at streams and springs is important for managing our water resources. Recent advancements in sensor technology allow us to get large amounts of nutrient data without the need to perform laboratory analysis for each data point. However, these data have not been integrated into numerical models that can be used to predict the pathways of contamination. In this study, we blend these two methods (sensing and modeling) and find that the large data sets generated by sensors can be used to improve numerical model predictions and reduce uncertainty associated with the model outputs. Future studies should consider how to best utilize large amounts of existing sensing data to mitigate contamination and improve water resources.

1. Introduction

Nitrate contamination of surface and ground water is ubiquitous around the world, degrades ecosystem health, and is projected to worsen over the coming century (Brooks et al., 2016; Burns et al., 2019; Wang et al., 2016). A primary reason for the degradation of water quality is the excessive application of fertilizer, manure, and sewage along flow paths that lead to streams, lakes, and estuaries (Kumar et al., 2018; Seitzinger et al., 2006). Deconvoluting a nitrate signal—to understand the source, location, and timing of contamination—is exceedingly difficult without data sets and models that represent the physics of transport on the timescales that they occur (Burns et al., 2019). One such landscape, with complex interactions of nitrate pathways, processes, and timing, is agricultural karst terrain, which is characterized by sinkholes, caves, and conduits that transport groundwater at surface water velocities (Al Aamery et al., 2021; Ford et al., 2019).

Karst terrain covers 7%–12% of the earth's land surface and is formed by the dissolution of soluble bedrock, such as limestone, making it a productive aquifer source that provides 25% of the world's population with water (Ford & Williams, 2007). As limestone dissolves, voids that span several orders of magnitude are created, ranging from micropores (~1 μm) to fractures (~1 mm) to conduits (~1 m) (White, 2002). These voids facilitate pathways of water transport that can broadly be defined as slow, intermediate, and quick, respectively (Fenton et al., 2017; Husic, Fox, Adams, Backus, et al., 2019). Various hydrologic tools have been used to identify the timing and magnitude of pathway delivery in karst. These tools include hydrograph separation (Ford et al., 2019), numerical modeling (Husic, Fox, Adams, Ford, et al., 2019), and high-frequency sensing data (Mellander et al., 2012). However, few studies to our knowledge have tested and extended the above concepts for nitrate transport in karst using nitrate sensor data (Schilling et al., 2019; Zhang et al., 2020) and none have utilized nitrate sensing together with numerical modeling to help constrain uncertainty regarding nitrate contamination. The increasing availability of high-frequency sensing data (Fenton et al., 2017; Schilling et al., 2019) and the improved computational capabilities of modeling packages (Husic et al., 2020) allows these knowledge gaps to be advanced at this time.

High-frequency sensors for the continuous in situ monitoring of nitrate have improved our understanding of nutrient export from karst environments (Opsahl et al., 2017; Schilling et al., 2019; Wang, Li, et al., 2020; Yue et al., 2019). Particularly, sensors allow us to collect data at time scales (<15 min) that are providing novel insight into the drivers of hydrology and biogeochemistry (Rode et al., 2016). Traditional data collection (e.g., weekly or monthly) may be suitable for characterizing general trends, but will often not capture temporally sensitive processes that may occur at the event-scale (Blaen et al., 2016, 2017). Nitrate sensing data can also be used in conjunction with other sensing data, such as specific conductance (SC), to gain greater insight to system function (Burns et al., 2019). Further, various indices can be derived from the discharge-concentration loops generated from high-frequency data, such as the Hysteresis Index (HI) and the Flushing Index (FI) (Jacobs et al., 2018; Lloyd et al., 2016a; Vaughan et al., 2017). The HI can be an indicator of the spatial proximity of a solute to an observation point or the relative speed by which a solute is connected to an outlet. In the case of clockwise hysteresis, the solute is proximally sourced or rapidly connected, whereas for counterclockwise hysteresis, the source is either distally located or slowly connected. On the other hand, the FI is an indicator of whether material transported causes an increase (mobilization) or decrease (dilution) in outlet concentration at peak discharge. While these indices and high-frequency data are useful for describing events that have already occurred, their integration into numerical modeling tools, which can be simulated into the future, is underdeveloped in the literature (Burns et al., 2019; Jensen & Ford, 2019).

Numerical modeling provides multiple advantages when used in conjunction with robust calibration data and field assessment, including the ability to forecast into the future, but also to provide estimates of processes that cannot be explicitly measured at the watershed scale, for example, soil-water infiltration (Al Aamery et al., 2021; Zhi et al., 2022). However, numerical models can be fraught with uncertainty, over-parameterization, and equifinality—the condition whereby many different (sometimes contradictory) parameter sets yield satisfactory model simulations (Beven, 2006). Fortunately, several methods can help mitigate these causes of error, including (a) collecting higher-resolution datasets with strict quality control measures, such as high-frequency sensing data; (b) testing multiple model structures and selecting the most parsimonious; and (c) integrating additional data, such as chemical tracers, and calibration objectives to reduce uncertainty bounds (Hartmann et al., 2013; Husic, Fox, Adams, Ford, et al., 2019). Calibrating model performance to discharge-concentration indices has recently been performed for grab-sampled data (Zhi & Li, 2020), but no studies to our knowledge have integrated high-frequency nitrate data and hysteresis indices to reduce modeled uncertainty. By narrowing broad parameterization to only include model realizations that accurately reflect hysteresis behavior, an improved simulation of water and nitrate routing and storage can be performed.

The objective of this study was to utilize high-frequency sensing data to advance our process understanding and model representation of nitrate transport. We collected 2-years of 15-min nitrate and SC data and performed hysteresis analysis for dozens of events to analyze source-timing dynamics. We used a suite of hydrodynamic and antecedent variables together with hysteresis results to develop a conceptual model of nitrate dynamics for a karst spring. Thereafter, we integrated high-frequency sensing data into a numerical model by using event discharge-concentration indices to provide additional parameterization constraints, improve process representation, and reduce model uncertainty. Lastly, we discuss how high frequency sensing data and numerical modeling can be coupled to inform land management and decision-making.

2. Study Site

The Royal Spring basin (58 km^2) is a limestone aquifer situated in the highly karstified Inner Bluegrass region of Kentucky, USA (Figure 1). The region is temperate with mean annual temperature and precipitation of

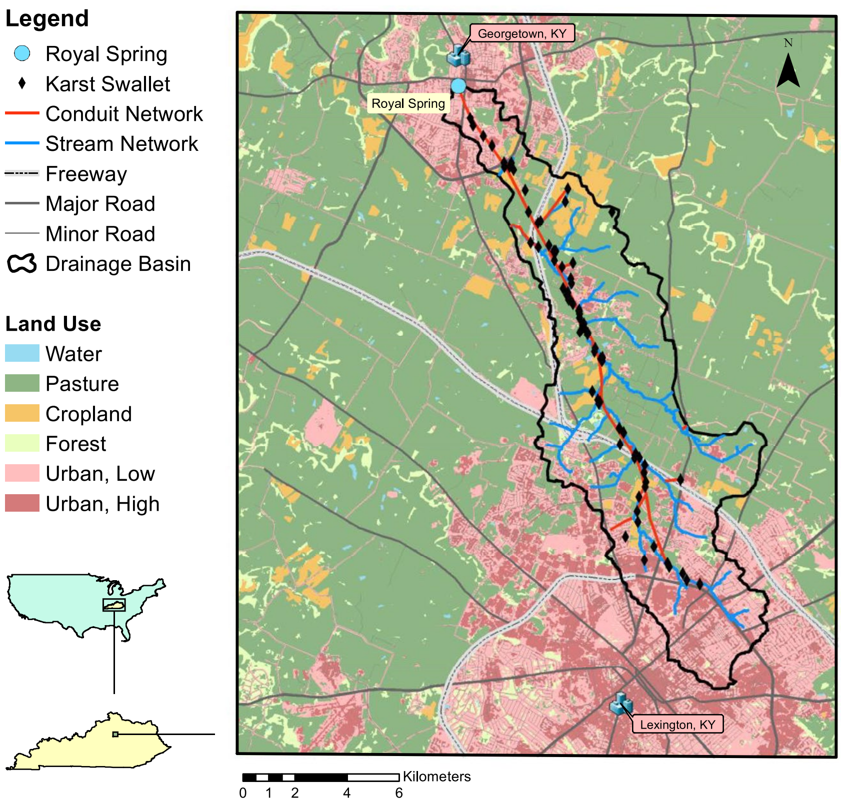

Figure 1. The location map of Royal Spring within Kentucky, USA. The surface stream and road networks are plotted alongside the karst swallets and inferred conduit network. Dominant land uses, including pasture, cropland, forest, low-intensity urban, and high-intensity urban are shown.

13.0 ± 0.7°C and 1,170 ± 200 mm, respectively (NOAA, 2020). Geology in the basin is composed of phosphatic limestone of the Middle Ordovician that is interbedded with shale (Spangler, 1982). Soils are moderately deep and well drained (NRCS, 2018). Land cover in the contributing drainage area is primarily agricultural (47% pasture and 7% cropland) with some forested areas (3%) and heavily urbanized headwaters (29% low urban development and 14% high urban development) (MRLC, 2019). The main stream that overlies the karst aquifer, Cane Run Creek, runs dry for over 80% of the year due to flow pirating through karst openings in the surface channel (Husic et al., 2017).

Royal Spring (243 m a.s.l; Q_{mean} = 0.67 $m^3\ s^{-1}$) is the largest spring in the Inner Bluegrass region and drains an extensive network of conduits, fractures, and matrix pores (Currens et al., 2015). The path of a dominant conduit network has been inferred from a series of dye traces and the mapping of dozens of in-stream sinkholes termed "swallets" (Paylor & Currens, 2004). Doppler sonar results from a series of groundwater wells that intersect the conduit network indicate that the main conduit is located 20 m below the earth's surface and has width of 6 m and height of 1 m (Husic et al., 2017). Mean measured velocities within the conduit are 0.12 $m\ s^{-1}$, allowing for rapid transport of material through the subsurface (Husic et al., 2017). Royal Spring experiences a wet season (December–May) when three-quarters of total discharge occurs. Lastly, Royal Spring serves as the principal water supply for the City of Georgetown and has recently experienced considerable water quality impairment, in part due to the high degree of surface-subsurface connectivity in the upstream basin (UKCAFE, 2011).

3. Methods

3.1. High-Frequency Sensing of Nitrate in Karst

3.1.1. Data Collection

In late 2017, a high-frequency sensor platform was installed directly downstream of Royal Spring, near a v-notch weir (USGS 03288110) where flow data have been recorded for 30 years. The platform includes a SUNA V2 ultraviolet nitrate analyzer as well as a YSI 6600 multiparameter sonde (Figure S1 in Supporting Information S1), both set to autonomously collect data at 15-min intervals. In this study, we use only the SC readings from the YSI 6600, although the other readings (dissolved oxygen, turbidity, temperature, and pH) were used to cross-check the data integrity of nitrate and SC readings (Clare, 2019). Weekly to-monthly grab samples were collected from Royal Spring for laboratory analysis of nitrate to validate sensor readings. Samples were filtered through a 0.45 μm syringe filter (Whatman 6780e2504) into a 40 mL borosilicate vial (I-Chem TB36-0040). Samples were immediately refrigerated, without preservation, and were analyzed within 48 hr. Field visits to Royal Spring were performed every 2 weeks and included retrieving stored data, replacing depleted batteries, and removing debris from the sensor platform. Sensor calibration was performed every other month as described in the respective sensor manuals. The SUNA V2 was removed from the site for annual factory calibration by Sea-Bird Coastal, which resulted in an 8-month data gap. Nonetheless, the periods totaling 16 months before and after the factory maintenance have minimal downtime and capture most events during the period. Meteorological data were retrieved from the nearby Bluegrass Airport station (NOAA USW00093820).

3.1.2. Data Analysis

Samples were analyzed in the Kentucky Geological Survey laboratory on a Dionex ICS-3000 ion chromatography system, consistent with U.S. EPA Method 300.0. The relationship between discrete data and sensor measurements was nearly linear with marginal error ($N_{\mathrm{lab}} = 0.94 \times N_{\mathrm{sensor}}$; $R^2 = 0.96$; $p < 0.05$; $n = 20$). Further, field duplicate samples ($n = 13$) had very low variability with a standard deviation of 0.03 mg N L^{-1}. With confidence in the sensor's ability to estimate nitrate concentration, we proceeded to assure the quality of the high-frequency time-series data through manual and automated procedures. We developed and implemented a flagging script, based on a publicly available code from Georgia Coastal Ecosystems LTER (Sheldon, 2015), to screen raw data and identify suspect data points. Questionable data, such as points far from the expected result but within the limitations, were manually included or excluded based on researcher judgment. For gaps created by rejecting data, linear interpolation was used if the gaps were less than 6 hr, otherwise gaps were left in place.

3.1.3. Event Identification and Likelihood of Extrema Timing

Qualifying storm events were extracted from the overall time-series using two criteria: data-availability and storm-intensity. First, only events with concurrent nitrate and discharge data availability were collated for analysis. Second, we manually defined the beginning of a storm as the point at which discharge begins to monotonically increase in response to rainfall. The end of an event was defined as the point where discharge reaches pre-storm conditions or a new event begins and disrupts the recession (Vaughan et al., 2017). A few events that have disrupted recessions were included in our analysis, which is a potential source of uncertainty in our statistical analysis. However, we include these events as their partial-hysteresis behavior can still be informative to model evaluation. In total, 34 qualifying events satisfied these criteria.

3.2. Hysteresis Analysis of Nitrate

3.2.1. Calculation of Discharge-Concentration Indices

Qualitative and quantitative assessments of nitrate and SC hysteresis were performed using high-frequency sensing data. Prior to any quantitative index calculations, a visual assessment was first performed for each storm to classify event hysteresis as clockwise (CW), counterclockwise (CCW), and figure-eight (F8). To quantify hysteresis, solute concentration (nitrate or specific conductance) and discharge were normalized by each event's respective maximum values to scale indices from −1 to +1. For each storm, the rising limb and falling limb were each divided into 50 equal-time intervals ($j = 1, 2, \ldots, 50$) that were used in calculations. A Savitzky-Golay digital filter was applied to discharge, nitrate, and SC time-series to reduce noise and to aid in visual presentation. Thereafter, the hysteresis index (HI) and the FI were calculated for each event.

The HI is a numerical indicator of hysteresis (Lloyd et al., 2016a) and was calculated by subtracting the normalized falling limb concentration from the rising limb concentration at each of the 50 intervals:

$$\mathrm{HI_i} = \mathrm{X_i^{RL}} - \mathrm{X_i^{FL}} \tag{1}$$

where $\mathrm{X_i^{RL}}$ is the normalized concentration during the rising limb at an interval and $\mathrm{X_i^{FL}}$ is the corresponding normalized concentration of the falling limb at the same interval. This procedure generates an array of HI values, and the mean of this array was used to represent overall storm hysteresis. Positive HI values indicate greater solute concentrations on the rising limb (clockwise hysteresis), which we interpret as a solute source that is rapidly connected to the outlet. Conversely, negative HI values indicate lower solute concentration on the rising limb (counterclockwise hysteresis), which we infer to mean a solute source that is slowly connected to the outlet. The magnitude of the HI indicates the strength of hysteresis (i.e., asynchronistic behavior) where "0" indicates a linear, chemostatic discharge-concentration relationship. The HI was calculated for both nitrate (HI_N) and SC (HI_S) data.

The FI is a numerical indicator of dilution or mobilization and was calculated as

$$FI = X_{\max Q} - X_{\text{initial}} \tag{2}$$

where $X_{\max Q}$ and X_{initial} are the normalized concentrations at peak and initial discharge, respectively. Positive FI values indicate an increase of concentration (i.e., mobilization) whereas negative FI values indicate a concentration decrease (i.e., dilution) along the rising limb. The magnitude of the FI indicates the intensity of flushing with a value of "0" representing no relationship between peak discharge and solute concentration. The FI was calculated for both nitrate (FI_N) and SC (FI_S) data.

3.2.2. Index and Event Categorization

Regarding nitrate, HI_N and FI_N analyses were performed on the original 15-min sensing data as well as daily averaged data. We daily averaged the nitrate data to allow for direct comparison to the numerical model, which runs at a daily timestep. Because daily averaging coarsens high-frequency data and results in the loss of detail, we employed the non-parametric Mann-Whitney test ($\alpha = 0.05$) to compare daily averaged versus 15-min index distributions. Daily averaged HI_N and FI_N were not significantly different from 15-min HI_N and FI_N ($p < 0.05$), providing us with confidence that information loss was minimal during the daily averaging process. However, it is important to note that coarsening data did smooth over some sub-daily figure-eight loops that occurred near peak discharges. In these instances, the event would no longer be visually classified as figure-eight, but the HI magnitude would remain relatively unchanged.

Modeled and observed storm results were compared with a bi-plot of HI_N and FI_N. HI_N classification is either rapid/clockwise connection (positive) or slow/counterclockwise connection (negative) whereas FI_N classification is either mobilization (positive) or dilution (negative). Therefore, four possibilities, or categories, arise for each storm event: rapid connection-peak flushing ($HI_N > 0$, $FI_N > 0$), slow connection-peak flushing ($HI_N < 0$, $FI_N > 0$), slow connection-peak dilution ($HI_N < 0$, $FI_N < 0$), and rapid connection-peak dilution ($HI_N > 0$, $FI_N < 0$). By analyzing the proportion of events falling within each category, we can develop an understanding of solute dynamics at this karst spring.

3.2.3. Statistical Analysis of Hysteresis Drivers

To determine potential drivers of nitrate delivery to Royal Spring, we compared a suite of hydrodynamic and antecedent variables to the hysteresis and nitrate timing classification of each event (Table S1 in Supporting Information S1). Hydrodynamic variables include P (total rainfall), Q_{peak} (peak flow), I_{flood} (flood intensity), t_{peak} (time-to-peak flow), and t_{storm} (storm duration). Antecedent variables include Q_{base} (baseflow discharge), AP_3 (3-day antecedent rainfall), AP_{14} (14-day antecedent rainfall), S_{base} (baseflow SC), and N_{base} (baseflow nitrate). We created a Pearson correlation matrix to assess how the magnitude of individual drivers may be correlated to the hysteresis and flushing indices for nitrate (HI_N and FI_N) and SC (HI_S and FI_S). A Pearson correlation coefficient (ρ) exceeding 0.5 was used as the threshold for determining strong correlation (Fovet et al., 2018). A Kruskal-Wallis test by ranks ($\alpha = 0.05$) was performed to assess if differences existed in the distribution of the predictor variables across hysteresis storm types, that is, figure-eight, clockwise, or counterclockwise. When a significant difference between groups was detected for a variable, a pairwise Tukey's honest significant difference test was conducted to establish which of the hysteresis types contributed to the difference.

We performed principal components analysis (PCA) to reduce the dimensionality of the data set and reveal underlying patterns among the variables. Hydrodynamic and antecedent drivers that are strongly related will have similar loading, that is, they plot close to one another on the PCA biplot. We infer the potential physical significance of the first two principal components (PCs) based on the position of the loadings along those PCs. Thereafter, we group the 34 events (observations) based on their hysteresis classification: clockwise, counterclockwise, and figure-eight. To visually assess how similar or dissimilar the subdivided groupings were, we generated confidence ellipses with a one standard deviation bound. Visual assessments of PCA biplots can be informative but also subjective (Skalski et al., 2018), thus we conduct a permutational multivariate analysis of variance (perMANOVA) test using the Euclidean distance between points as the dissimilarity measure (Anderson, 2017). Rejecting the perMANOVA null hypothesis indicates that the centroid and dispersion are significantly different between groups.

3.3. Reducing Numerical Model Uncertainty

3.3.1. Model Description

To model hysteresis behavior, we use a calibrated hydrologic and nitrogen model of Royal Spring, which simulates daily discharge as well as the nitrate, ammonium, and dissolved organic nitrogen concentrations (Husic, Fox, Adams, Ford, et al., 2019; Figure S2 in Supporting Information S1). In short, the lumped numerical model utilizes a series of cascading conceptual reservoirs to represent the transfer and storage of water and nitrate in conduit, soil, epikarst, and phreatic zones. Over 30 million uncertainty simulations were performed, and five alternative model structures tested, yielding a robust assessment of equifinality and 1,382 successful model parameterizations. The important parameters identified in the original analysis are listed in Table S2 in Supporting Information S1 along with their calibrated values and variabilities. Two changes to the original model structure presented in Husic and others (2019b) were made for improved performance in this study. First, we changed the potential evapotranspiration module from Penman-Monteith to McGuinness-Bordne as the latter is more appropriate for lumped rainfall-runoff models (Oudin et al., 2005). Second, nitrate recharge in the original model had a discrete seasonal-value, which we now transform into a continuous daily value by fitting a Fourier series to the seasonal estimates. To test model performance with grab-sample data, we project the historic model run (October 2012 to October 2016) forward to June 2019 and find good agreement between projected model results and subsequently collected nitrate data (Figure S3 in Supporting Information S1).

Adding hysteresis as a calibration target for the model has the potential to reduce equifinality, the process by which differing model parameterizations give equivalent results (Beven, 2006), by constraining the uncertainty of modeled processes. The two primary processes that impact modeled concentration and thus hysteresis are the routing and storage of nitrate. In the first process, spring discharge is sourced from three pathways (quick, intermediate, and slow) that broadly represent flow routed across three porosities in mature karst (conduit, fracture, and matrix, respectively). In the second process, spring nitrate is altered by the residence time and reactivity of nitrate in aquifer storage zones (soil, epikarst, and saturated matrix). The dominant routing mechanisms (and the parameters that represent them, Table S2 in Supporting Information S1) include fraction of rainfall as concentrated recharge (X), soil (k_{soil}) and epikarst (k_{EL}) percolation coefficients, and the activation of highly connected soil ($\text{V}_{\text{S,MAX}}$) and epikarst ($V_{\text{E,FAST}}$) pathways. The dominant storage and reactivity mechanisms (and the parameters that represent them, Table S2 in Supporting Information S1) include soil field capacity ($V_{\text{S,MIN}}$), aquifer pumping rate (Q_{PUMP}), first-order rate constants (k_{DEN}, k_{NITR}, k_{MIN}), and seasonal nitrate recharge concentrations ($\text{C}_{\text{NO}_3^-\text{(F)}}$, $\text{C}_{\text{NO}_3^-\text{(W)}}$, $\text{C}_{\text{NO}_3^-\text{(SP)}}$, and $\text{C}_{\text{NO}_3^-\text{(SU)}}$).

3.3.2. Hysteresis Evaluation and Uncertainty Reduction

The primary evaluation objective was to compare modeled and observed HI_N and FI_N behavior (Figure 2). Data-observed and model-simulated events with the same categorization (e.g., rapid connection-peak flushing, slow connection-peak flushing, slow connection-peak dilution, and rapid connection-peak dilution) passed the behavioral test whereas differing categorizations failed the test. Visual inspection of the modeled hysteresis plots was also conducted to verify accurate event characterization. Each parameterization of the model (of the 1,382 total parameter sets) generated HI_N and FI_N results for all events (Figure 2). As there presently does not exist a standard method for evaluating hysteresis in numerical models (Mahoney et al., 2020), we surmised that a model parameterization should at least predict a majority of events if it is behavioral. Thus, we retained only the parameterizations that correctly categorized more than 50% of the chemodynamic events (n = 26). Chemodynamic events were defined as those with FI_N and HI_N magnitudes of at least 0.10 (i.e., observable hysteresis or flushing; Bieroza et al., 2018). This distinction allowed for more accurate inclusion of only events that informed hysteresis processes. Thereafter, the remaining parameterizations were used to generate new hysteresis-improved nitrate prediction bounds and parameter variabilities. Overall prediction

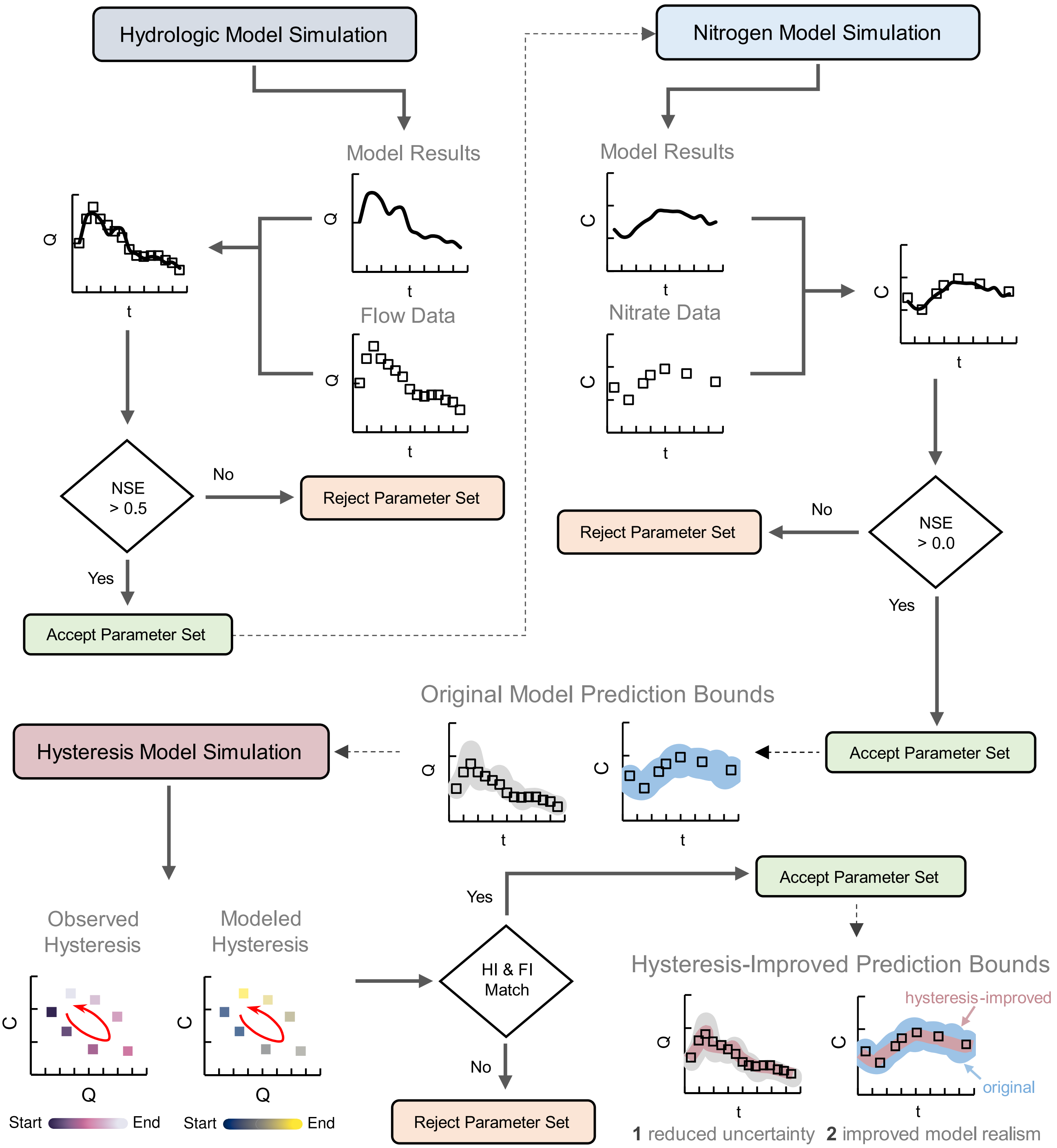

Figure 2. Nitrate hysteresis model evaluation flow chart. Initial hydrologic and nitrogen model results are from Husic, Fox, Adams, Ford, et al. (2019) and represent simulations calibrated to grab-sample data. Hysteresis-improved prediction bounds reflect the reduction in overall model uncertainty by satisfying hysteresis evaluation criteria. The Q, C, and NSE abbreviations represent discharge, nitrate concentration, and the Nash-Sutcliffe Efficiency. The term “parameter set” refers to possible combinations of hydrologic and nitrogen model parameters in a given simulation.

uncertainty reduction was calculated as the average percent minimization of the 95% prediction interval of the hysteresis-improved bounds compared to the original model bounds. Likewise, parameter variability reduction was calculated as the percent change in original parameter variability width with hysteresis-improved variability.

4. Results and Discussion

4.1. High-Frequency Sensing and Hysteresis of Nitrate in Karst

4.1.1. High-Frequency Sensing of Solute Timing

We collected 2 years of 15-min sensing data and observed both seasonal and sub-daily variations in nitrate and SC (Figure 3). The study period was particularly wet (1,626 ± 286 mm yr^{-1}) compared to the historical rainfall average (1,170 ± 200 mm yr^{-1}). Due to wet conditions, baseflow was maintained at an elevated level without no-flow summer periods, which are characteristic of most years. Specific conductance values ranged between 247 and 611 μS cm^{-1} and had an average value of 435 ± 62 μS cm^{-1}. Baseflow SC values were typically greater during the dry summer and lower during the wet winter. Mean nitrate concentrations (2.98 ± 0.70 mg N L^{-1}) were well into the hypereutrophic range (>1.5 mg N L^{-1}) but were always below the USEPA maximum contaminant level (10 mg N L^{-1}) for drinking water. At the seasonal timescale, nitrate concentrations ranged from 1 to 5 mg N L^{-1} with peaks in the winter and troughs in the summer.

High-frequency sensing captured the nitrate and SC responses of 34 events over the 2-year study period (Figure 4). At the sub-daily timescale, every storm event substantially altered nitrate and SC with a few trends emerging across storms. Dilution of both nitrate and SC commonly occurred in response to aquifer recharge, reflecting the arrival of dilute stormwater. In several events, the dilution behavior of the two solutes nearly mirrored one another (see Events 2, 5, and 6 as examples). Exceptions to the typical solute dilution behavior, such as the observed nitrate mobilization in Event 9, are attributable to wet antecedent conditions whereby the influence of the prior event (Event 8) had not entirely subsided, thus seeding initially diluted nitrate levels that are later increased by the arrival of nitrate-rich water. Another exception is Event 25 where SC increases, rather than decreases, during the rising limb of an event while nitrate follows dilution behavior. This event occurred during the winter season where salt applied to roadways may be routed into the subsurface. After the initial dilution of nitrate and SC by the arrival of quick flow, solute behavior during the hydrograph recession was more variable. For example, SC often returned to pre-event levels (see Events 16, 20, and 29) as does nitrate (see Events 2, 6, and 30). However, in several events the nitrate concentration during hydrograph recession exceeded the initial baseflow nitrate (see Events 3, 4, 7, 23, and 32), indicating slow connection of a potentially distal source.

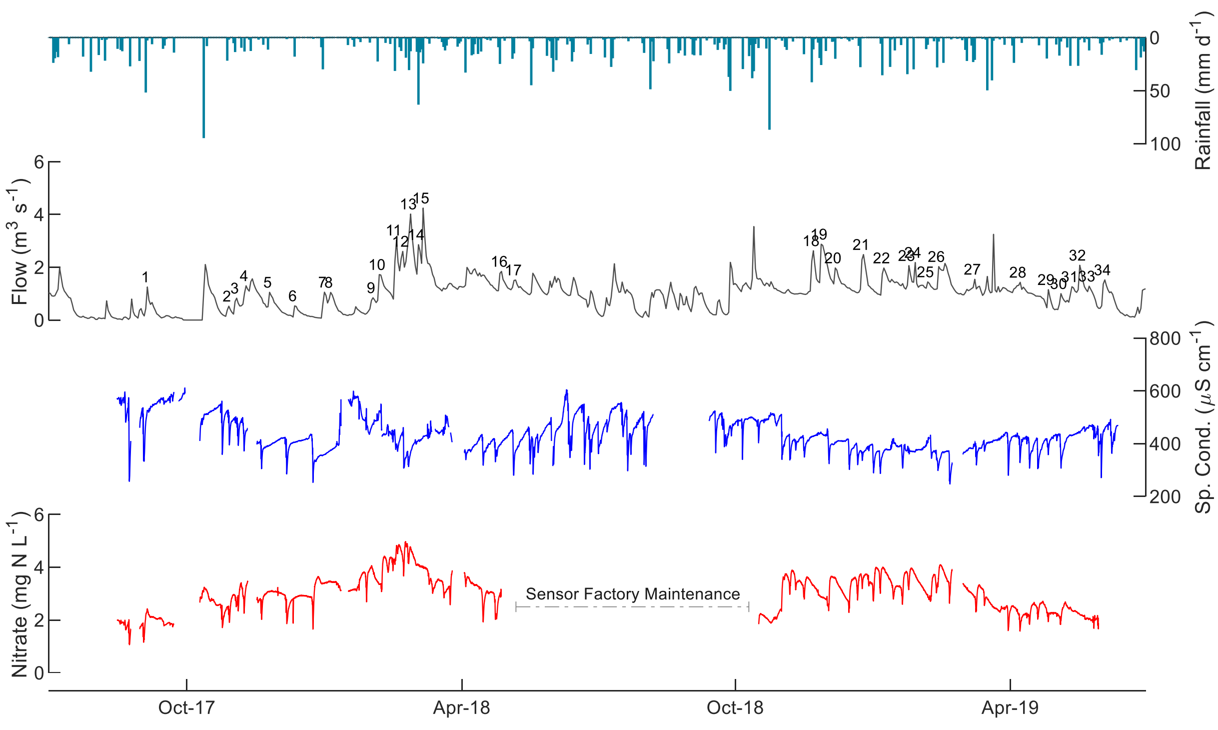

Figure 3. Daily rainfall, daily average flow, 15-min specific conductance, and 15-min min nitrate data from Royal Spring. Qualifying events are labeled at their respective peak discharges.

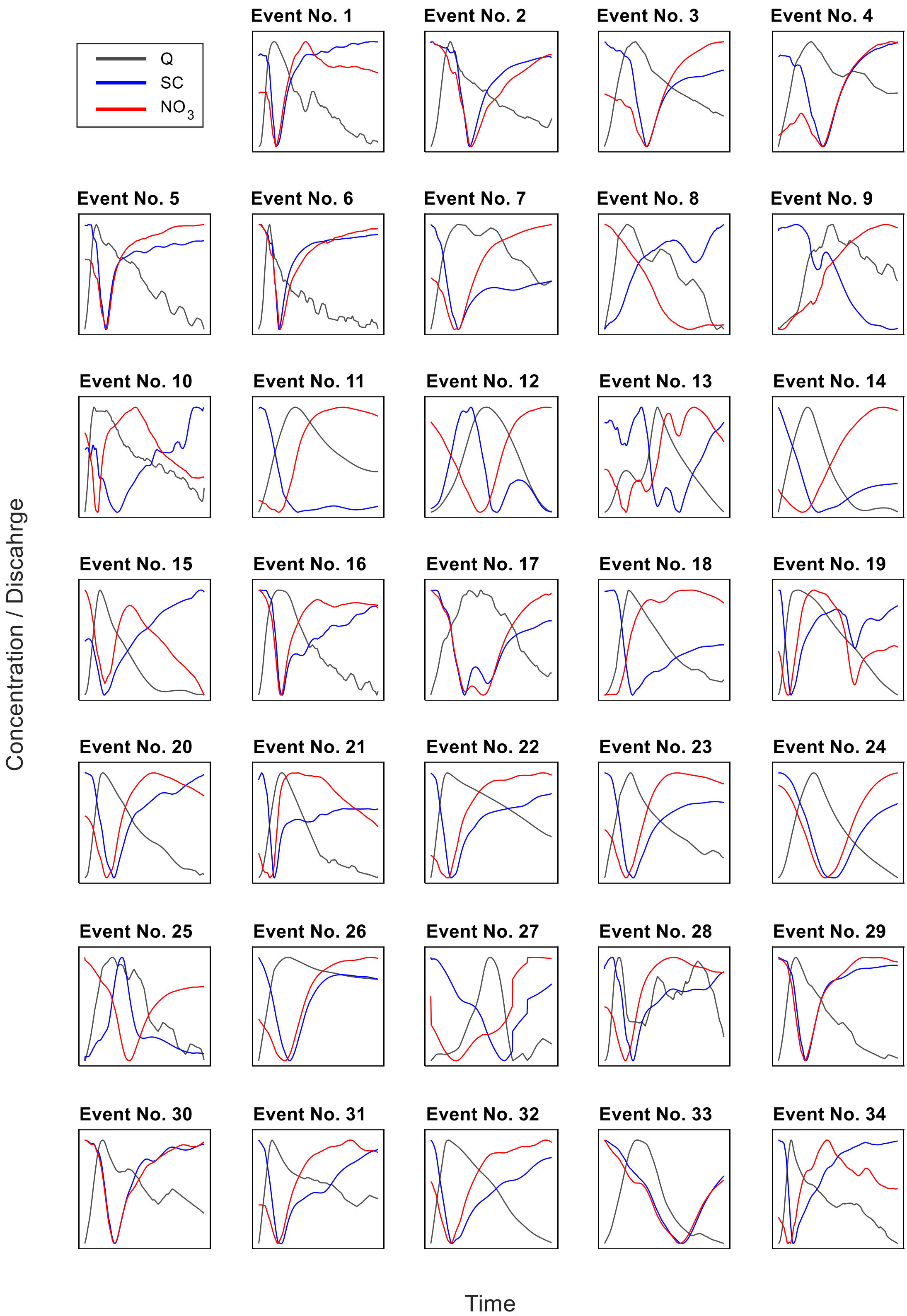

Figure 4. Storm discharge (*Q*) and 15-min nitrate (NO_3) and specific conductance (SC) time-series for qualifying events. For each subplot, the *x* axes are normalized by event duration and the 2 *y* axes are normalized by respective event discharge and concentration. Refer to Figure 1 for event dates and magnitudes. Note: jagged behavior during some flow recessions was caused by pumping of spring water from directly upstream of the monitoring gage.

The recent growing body of nitrate sensing literature in karst settings reinforces the concept of karst springs as dynamic integrators of storm-event dynamics. Complex contributions of quick and slow flow can cause nitrate dilution and mobilization within the same event (Huebsch et al., 2014), which are tied to the underlying maturity of the karst structure (Schilling et al., 2019). Schilling et al. (2019) showed, using sensing data, that in an immature karst system in Iowa, USA, nitrate stored in overburden material can provide a constant source of high concentration material whereas in a neighboring mature system event recharge via sinkholes significantly diluted spring concentrations. The nitrate response in Royal Spring both resembles the mature system, where dilution is common, but also the immature systems where events are characterized by mobilization of nitrate during storm recession. The similarity in nitrate response patterns occurs despite concentrations that are 4–6 times greater in the Iowa springs, which are a result of more extensive cropping in the Iowa systems (>70%) compared to Royal Spring (7%). While Schilling and others (2019) do not present discharge or SC data, their results—in addition to ours—fit a conceptual model whereby elevated nitrogen loading is largely sourced from the soil and epikarst. Further credence to the idea of substantial contributions from soil and near-surface epikarst zones during peak nitrate concentrations is provided by modeling studies across a range of settings, which show large epikarst flow contributions, ranging from 49% to 83%, during flood recession (Husic, Fox, Adams, Ford, et al., 2019; Wang, Chen, et al., 2020). Thus, with an understanding of our time-series dynamics in the context of other high-frequency sensing studies in karst, we proceeded to systematically investigate the asynchronous behavior between discharge, SC, and nitrate.

4.1.2. Hysteresis Classification of Storm Dynamics

Visual assessment of SC hysteresis loops shows single-shape dominance (26 out of 34 events are clockwise) (Figure 5). On the other hand, nitrate hysteresis was more variable with no single shape dominating the signal: $n = 11$ for clockwise, $n = 13$ for counterclockwise, and $n = 10$ for figure-eight (Figure 6). A quantitative inspection of event categorization using the hysteresis (HI_S and HI_N) and flushing (FI_S and FI_N) indices for SC and nitrate, respectively, showed similar patterns to visual assessment. Regarding SC, 27 of 34 events experienced rapid connection-peak dilution ($HI_S > 0$, $FI_S < 0$), suggesting that initial high SC (~550 μS cm^{-1}) was diluted immediately by arriving waters. Regarding nitrate, 15 events showed rapid connection-peak dilution ($HI_N > 0$, $FI_N < 0$) while 14 events showed slow connection-peak dilution ($HI_N < 0$, $FI_N < 0$). Of the remaining five non-dilution events (Events 9, 11, 18, 19, and 21), all were characterized by slow connection-peak flushing. No events were characterized by rapid connection-peak flushing, suggesting that high nitrate concentrations typically arrived only after sufficient activation of slower pathways and/or distal sources. It is important to note that many events with "dilution" during peak discharge, often had nitrate maxima that occurred during hydrograph recession (examples include Events 22 and 23). Thus, many events may dilute on the rising limb, but mobilize on the falling limb. Taken together, the high degree of variability in the nitrate hysteresis results suggest a dynamic system whose response is likely influenced by the hydrodynamic and antecedent conditions at the time of the event.

The high variability in hysteresis shape in our karst spring differs somewhat from high-frequency measurements reported in streams where hysteresis behavior is largely controlled by a single shape (Aguilera & Melack, 2018; Jacobs et al., 2018; Kincaid et al., 2020; Lloyd et al., 2016b; Vaughan et al., 2017; Williams et al., 2018), but with some exceptions (Baker & Showers, 2019; Duncan et al., 2017). For example, in two rural sites in the United Kingdom, Lloyd et al. (2016a) showed that 95% of events showed clockwise dilution of nitrate. Likewise, in more crop-dominated sites in the United States, Williams et al. (2018) reported that 74% of events exhibited anti-clockwise sourcing of nitrate, a process largely associated with rises in the water table connecting nitrate storages and drainage via artificial tiles. In an urban stream with mostly clockwise hysteresis (65% of events), Duncan et al. (2017) showed that events transitioned from counterclockwise to clockwise after a certain storm intensity threshold was exceeded (i.e., $Q_{peak} > 20 \times Q_{base}$), suggesting that small events sourced distal nitrate whereas large events, dominated by new water, sourced proximal nitrate. Interestingly, figure-eight patterns are less commonly observed in streams, constituting around 10% or less of multiple rural (Williams et al., 2018) and urban (Duncan et al., 2017) system responses. The only study we found that uses high-frequency sensing and hysteresis indices at a karst spring was by Zhang et al. (2020), and they showed similarly varied nitrate response with three counterclockwise, eight figure-eight, and four unclassified events. With an understanding of the context of our hysteresis results within the existing literature, we investigate the potential hydrodynamic and antecedent drivers of the variable nitrate response observed at Royal Spring.

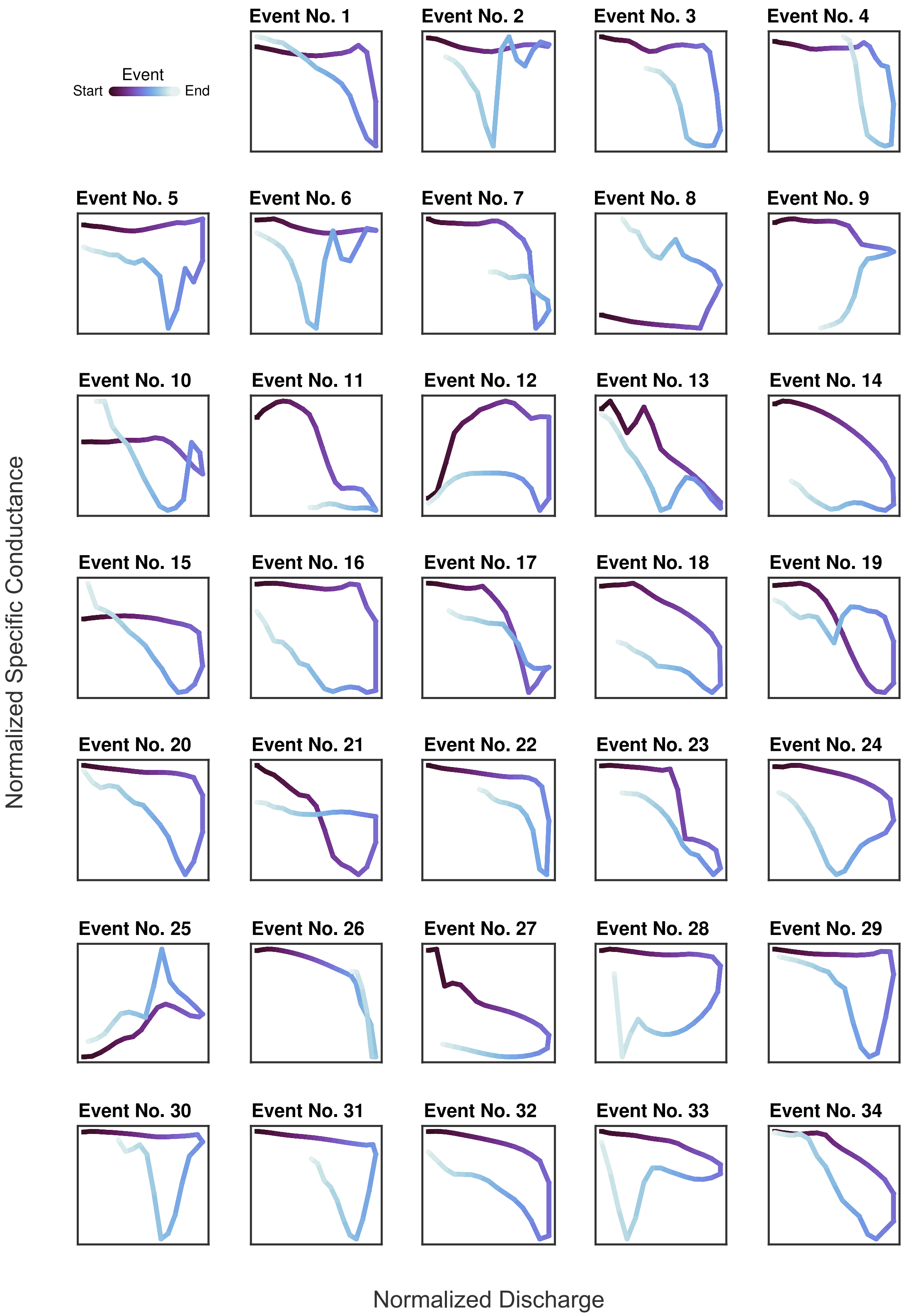

Figure 5. Conductivity hysteresis analysis of storm events as calculated using 15-min sensor data. Hysteresis is color-coded from start to the end of an event. The *x* and *y* axes for each subplot are normalized by peak event discharge and concentration, respectively.

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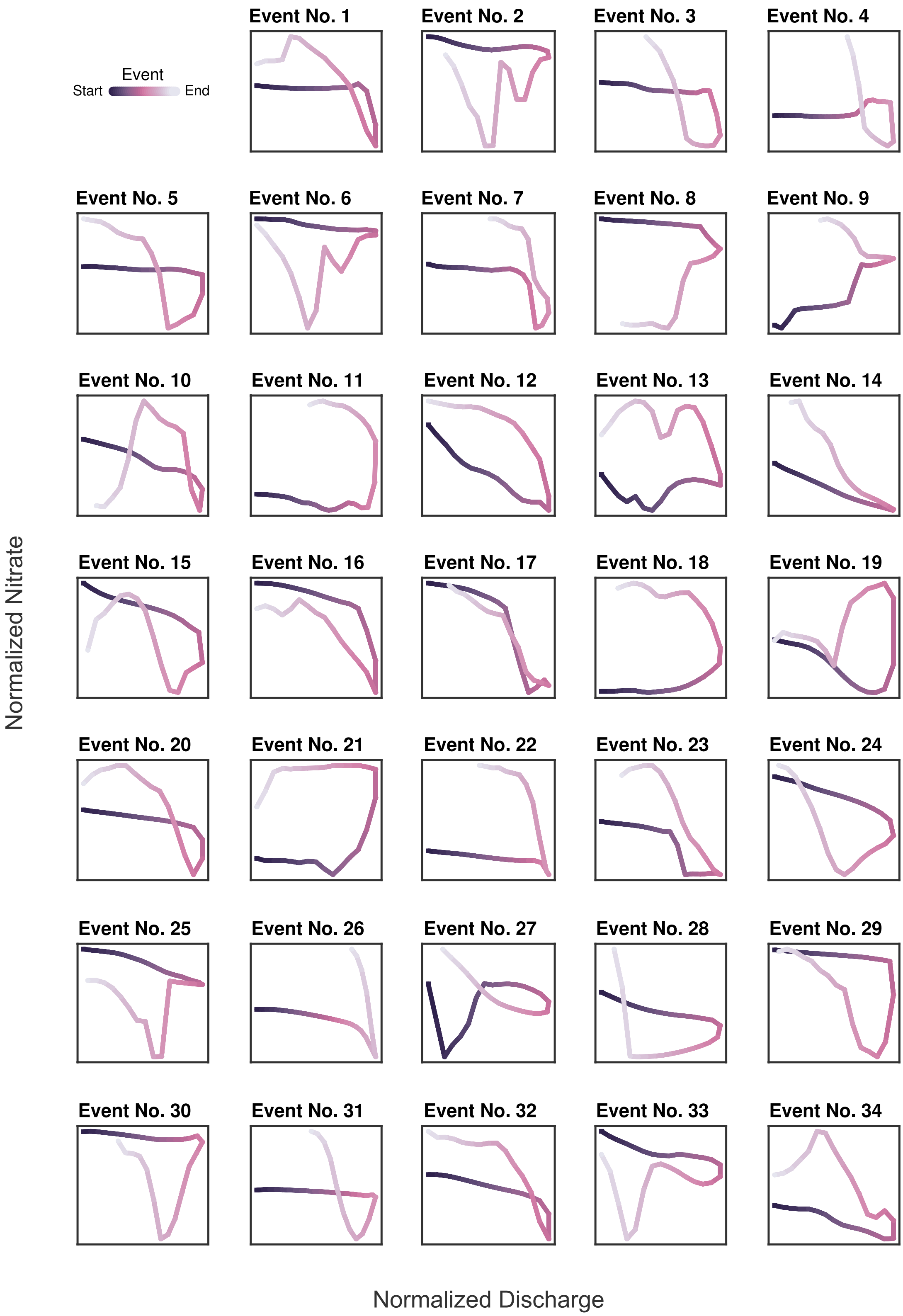

Figure 6. Nitrate hysteresis analysis of storm events as calculated using 15-min sensor data. Hysteresis is color-coded from start to the end of an event. The *x* and *y* axes for each subplot are normalized by peak event discharge and concentration, respectively.

4.1.3. Conceptual Model of Nitrate Transport Drivers in Karst

We develop a conceptual model to explain the observed nitrate hysteresis responses at the karst spring using a suite of hydrodynamic (P_{storm}, I_{flood}, Q_{peak}, t_{peak}, t_{storm}) and antecedent (Q_{base}, AP_3, AP_{14}, S_{base}, N_{base}) explanatory variables. Strong correlation ($\rho > |0.5|$) was not observed between FI_N and HI_N and any of the explanatory variables (Figure S4 in Supporting Information S1). However, several variables did show moderate correlation ($|0.3| > \rho > |0.5|$) with FI_N, including +0.44 with S_{base} and −0.36 with AP_{14}. Regarding HI_N, the strongest correlation was −0.49 with P_{storm} and −0.47 with $Q_{peak.}$ An analysis of variance showed that the only variables that significantly differed between hysteresis types were P_{storm} and Q_{peak} ($p < 0.05$). Tukey's test showed that counterclockwise hysteresis, that is, slow connection of nitrate sources, events had significantly larger values of P_{storm} and Q_{peak} than did clockwise events ($p < 0.05$). While these results indicate a few important variables, they unsurprisingly suggest that the nitrate hysteresis behavior at the spring does not express a linear response to any individual driver, but rather behavior at the spring likely depends on a combination of hydrodynamic and antecedent drivers.

Principal components analysis results helped to reduce the dimensionality and reveal underlying relations in the data; the first two PCs explained 57% of overall variance (Figure 7). PC 1 has positive association with all variables apart from S_{base}, and we broadly interpret it to represent storm size (small vs. large). On the other hand, PC 2 has positive association with P_{storm}, I_{flood}, Q_{peak}, and S_{base} while it has negative association with AP_3, AP_{14}, and N_{base}. We broadly interpret PC 2 to represent hydrologic connectivity (antecedent vs. flashy). Events in positive PC 2 space are connected by hydrodynamic, event-scale indicators (P_{storm}, I_{flood}, and Q_{peak}). Positive association of PC 2 with S_{base} indicates events in this space have high pre-event SC, which typically happens during prolonged baseflow periods where connectivity to non-baseflow water storages has ceased. Taken together, the positive PC 2 space broadly represents "*flashy connectivity*" that is driven more by event hydrodynamics than pre-existing antecedent conditions. On the other hand, events in negative PC 2 space are connected by longer-term wetness indicators: AP_3, AP_{14}, and N_{base} (N is seasonally high during the wetter winter). S_{base} is inversely related to long-term indicators because SC becomes smaller as antecedent wetness increases. Overall, the negative PC 2 space represents "*antecedent connectivity*" that is related more to antecedent wetness than hydrodynamic event forcing.

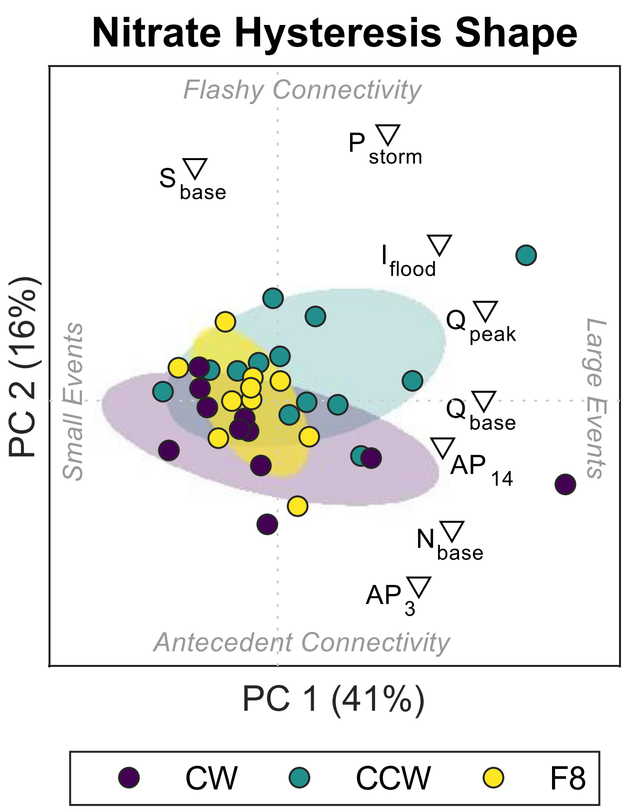

Figure 7. Principal components analysis biplots showing observations (filled circles) and hydrologic variable loadings (open triangles). Observations (events) are grouped based on nitrate hysteresis type and are encircled by a confidence ellipse. The first PC broadly represents event size (small vs. large) whereas the second PC broadly represents connectivity (flashy vs. antecedent). Dotted lines are drawn to show the *x*- and *y*-axes origins.

Regarding nitrate hysteresis, the clockwise and counterclockwise event clusters plot distinctly from one another on the PCA biplot (Figure 7). The figure-eight event grouping appears to be in an intermediate position between the other two clusters. A perMANOVA test confirms the visual assessment and indicates that significant differences ($p < 0.05$) exist between clockwise and counterclockwise observations. With significant groupings identified, we next investigate the potential drivers and processes that cause certain hysteresis types. Looking at PC 1, we see that both clockwise and counterclockwise event types are associated with storms large and small, but that clockwise events occupy a slightly more negative space along PC 1, suggesting greater alignment of clockwise (rapid connection) events with smaller storms. However, PC 1, which we infer to represent storm size, does not on its own explain differences in our hysteresis classifications. On the other hand, we find greater separation in clockwise and counterclockwise observations along PC 2, which indicates antecedent connectivity of nitrate may inform hysteresis type. In summary, the clockwise hysteresis (rapid connection) ellipse trends toward smaller events and antecedent connectivity whereas the counterclockwise (slow connection) ellipse trends toward larger events and flashier connectivity.

The major difference we observe between the drivers of hysteresis in our spring versus stream systems is in the relative importance of antecedent conditions. In surface streams, storm size has been highlighted as a major driver of hysteresis type (e.g., Duncan et al., 2017; Jacobs et al., 2018), with both studies identifying a "tipping point" in storm intensity after which storm hysteresis transitions from counterclockwise to clockwise. Antecedent conditions are seen as of lesser (Jacobs et al., 2018) or no (Duncan et al., 2017) importance relative to storm intensity. This discrepancy may result from a shift in behavior from transport-limited to supply limited in surface streams. At low flows, the proximal nitrate sources are large enough that limited flow rates can continually transport nitrate. However, as flows increase, the nitrate source zones may become exhausted at which point the stream transitions to clockwise behavior. The impact of flow intensity is mitigated in karst springs because they typically have a maximum-discharge capacity (Bonacci, 2001; Husic et al., 2017). Once this capacity is reached, any additional loading into the subsurface conduit network finds other energetically favorable pathways to traverse, such as overtopping sinkholes and remaining in the surface channel, overflowing to intermittent springs, or returning to aquifer storage as conduit-matrix exchange (Bettel et al., 2022; Bonacci, 2001). The net impact is that the karst system can continually export nitrate-storages at a consistent rate without source exhaustion that may be caused by extreme events. Thus, how well the connection between nitrate storage has already been established through antecedent connectivity determines system response as opposed to storm size.

We glean new insights into the physical significance of nitrate hysteresis in karst from a variety of sensing and statistical methods. Events exhibiting clockwise hysteresis are linked to greater antecedent connectivity whereas events with counterclockwise hysteresis align with flashier connectivity (Figure 7). First, events with counterclockwise hysteresis are initially disconnected from nitrate-rich reservoirs in the basin, which for our system are the soil and epikarst (Husic, Fox, Adams, Backus, et al., 2019). Upon establishment of connectivity through flashy hydrodynamic forcing (large P_{storm} and I_{flood}), nitrate-rich water arrives in the latter end of the hydrograph. This creates the counterclockwise hysteresis pattern, which is inferred as either distal sourcing or slow connection of a source. On the other hand, events with clockwise hysteresis already have antecedent connectivity to near-surface, nitrate-rich storages at storm onset. This is indicated in part by lower S_{base} ("fresher" water) and higher N_{base}, respectively. Because of this antecedent connectivity, new water arrival, sourced primarily from runoff diverted into sinkholes, acts to dilute nitrate concentrations during rapid contribution. As dilute quick flow subsides, connectivity to pre-event nitrate-rich stores is reestablished as the primary source of solutes to the spring. Events with figure-eight hysteresis appear to be a middle-ground of the flashy and antecedent connectivity end-members, but we are not able to fully resolve the direct causes of figure-eight hysteresis with our current data. Further, our interpretations of nitrate hysteresis are limited to the temporal domain as we lack the necessary spatial data (e.g., groundwater, sinkholes, and soil) to infer proximal or distal connectivity of nitrate sources.

Our concept model for karst nitrate dynamics shows some corroboration with the numerical model structure (Figure S2 in Supporting Information S1), the karst literature, and previous work in this karst basin. The behavior of water and nitrate reflect well the quick-flow, soil and epikarst reservoirs (intermediate-flow), and phreatic zone (slow-flow) in Figure S2 in Supporting Information S1. These concepts agree with often cited interpretations for triple-transfer (i.e., quick-, intermediate-, and slow-flow) pathways in karst studies (e.g., Chang et al., 2019; Fenton et al., 2017; Ford et al., 2019). Prior work in this basin, using water isotopes ($\delta 2\mathrm{H}_{\mathrm{H_2O}}$ and $\delta^{1}8\mathrm{O}_{\mathrm{H_2O}}$) and rainfall-discharge cross-covariance analysis, also showed that spring discharge responds to rainfall input on the same day that it occurs (Husic, Fox, Adams, Backus, et al., 2019; Husic, Fox, Adams, Ford, et al., 2019). Further, high-resolution sampling of storms events in the basin for nitrate stable isotopes ($\delta^{1}5\mathrm{N}_{\mathrm{NO_3}}$ and $\delta^{1}8\mathrm{O}_{\mathrm{NO_3}}$) showed that maximum nitrate concentration lags peak spring discharge (Husic, Fox, Adams, Ford, et al., 2019) and likely originates from the soil and epikarst zones (Husic et al., 2020). Other studies also corroborate the idea that the greatest nitrate concentrations may be reflective of a soil-nitrate pool that becomes highly connected to the epikarst during wet conditions (Zhang et al., 2020). With confidence in the conceptual model of nitrate karst dynamics, we proceeded to improve numerical model representation by integrating this new sensor-derived knowledge into model evaluation.

4.2. Numerical Modeling and Reducing Model Uncertainty

4.2.1. Modeling Nitrate Hysteresis

Comparison of model-simulated and data-observed hysteresis showed overall agreement (Figure 8). The best performing model simulation (of the 1,382 total realizations) correctly characterized both HI_{N} and FI_{N} for 23 of the 34 events and at least one of HI_{N} or FI_{N} for 30 of the 34 events. Two modeled events wrongly predicted both HI_{N} and FI_{N}, and a further 2 events modeled no hysteresis at all, that is, monotonically increasing or decreasing flow during an entire event. The numerical model was most accurate in characterizing events with dilution

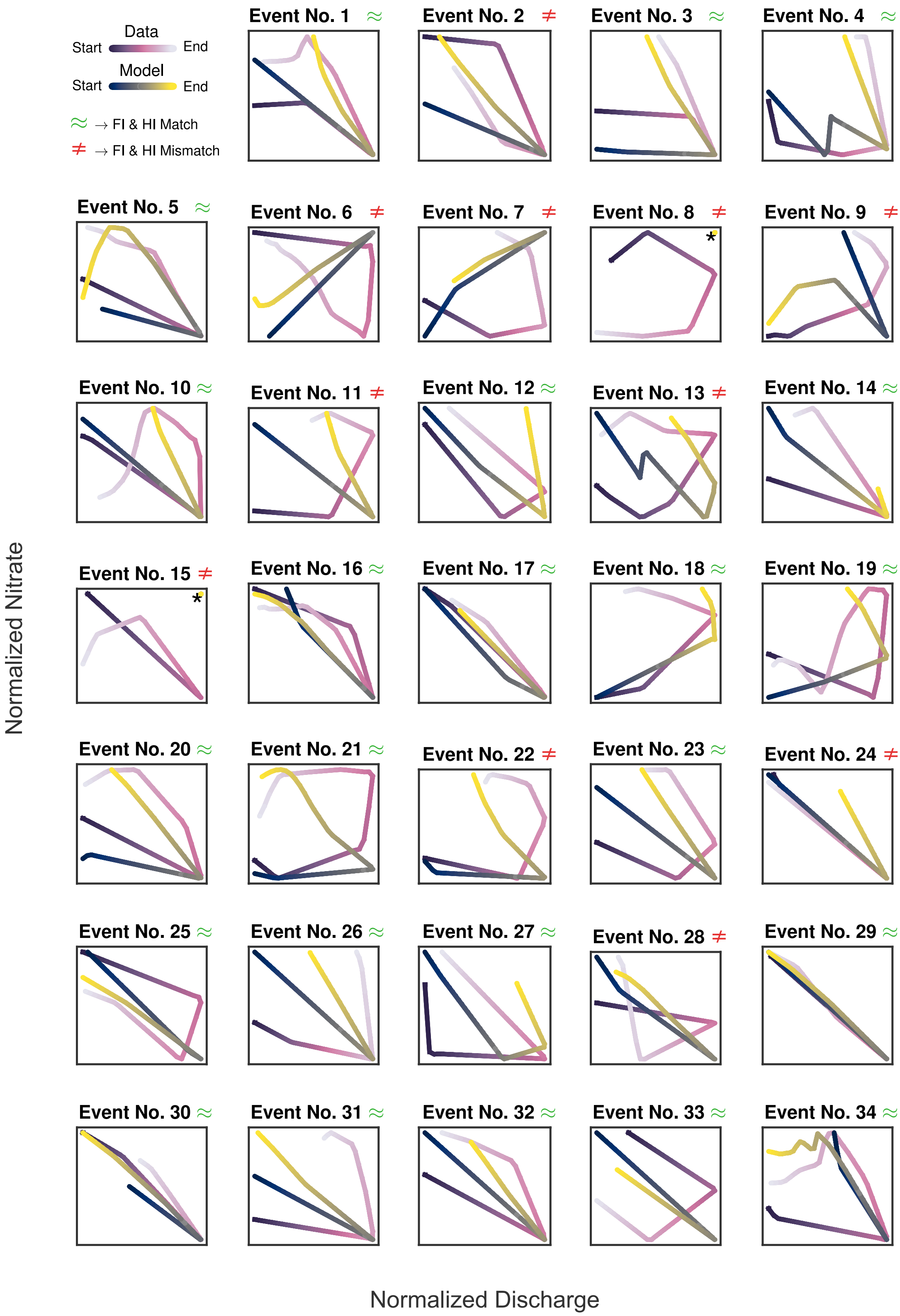

Figure 8. Model-simulated nitrate hysteresis compared to hysteresis calculated from daily averaged sensor data. An approximately equal symbol (≈) indicates that modeled and observed HI_N and FI_N have similar behavior, whereas a not-equal symbol (≠) indicates that modeled and observed HI_N and FI_N differ. An asterisk (*) inset within a subplot indicates no hysteresis predicted by model. The *x* and *y* axes for each subplot are normalized by peak event discharge and concentration, respectively.

($FI_N < 0$) at peak discharge (23 correct of 27 observed diluting events) and where nitrate connection to the spring occurred slowly ($HI_N < 0$; 24 correct of 26 observed slow connection events). The less-common event types, including flushing during peak discharge ($FI_N > 0$; 3 correct of 7 total) and rapid connection of nitrate ($HI_N > 0$; 3 correct of 8 total) were not characterized as accurately. Near-chemostatic events (e.g., Event 24) can be correctly or incorrectly characterized with slight differences in clockwise or counterclockwise rotation.

Mean HI_N and FI_N statistics reinforced the visual assessment with model-simulated HI_N and FI_N of -0.21 ± 0.18 and -0.54 ± 0.56, respectively, compared to the data-observed HI_N and FI_N of -0.26 ± 0.42 and -0.30 ± 0.55, respectively (Figure 9). Statistical testing shows that the modeled and observed HI_N distributions were not significantly different ($p > 0.05$) whereas FI_N distributions showed some differences ($p = 0.047$). Nonetheless, the generally positive model results give us confidence to further investigate and infer connectivity of nitrate source-zones and flow pathways in our karst spring.

Perhaps unsurprisingly, the model did better at matching events that fit the underlying conceptualization that was the basis for the numerical model (Figure S2 in Supporting Information S1). However, despite the general success in matching observed hysteresis, it is informative to investigate the types of events the model failed to reproduce. In particular, while the model captures the most common event type—initial dilution (27 of 34 events) followed by slow connection of nitrate (26 of 34 events)—it generally is less successful simulating rapid nitrate connection followed by peak-dilution (lower right quadrant of Figure 9). In rapidly connected events, such as Event 8 which began only 10 hours after the conclusion of Event 7, nitrate concentrations remain high on the rising limb and drop during the falling limb. The wet antecedent conditions may have caused runoff generated during this event to remain in the surface stream and avoid being abstracted into the conduit network by sinkholes, mitigating the immediate diluting influence of quick flow. However, in our numerical model conceptualization, quickflow always has same-day delivery to the spring (Figure S2 in Supporting Information S1), bypassing any potential attenuation in the surface. Future improvements to our lumped model could consider the antecedent conditions in the surface stream in determining the fraction of runoff abstracted to the conduit network. Fully distributed reactive transport models can explicitly consider these processes (Zhi et al., 2022), but the additional data, parameterization, and computational requirements may not warrant such models, particularly if an existing lumped model performs well in most cases—as for our system.

Numerically simulating hysteresis is a promising area for improving process representation of our models. Studies that do this are rare in the literature as high-frequency sensing of nitrate is an expensive upfront cost and is limited to fewer sites than traditional grab sampling (Burns et al., 2019) and effective numerical models require extensive knowledge of the study site (Hartmann et al., 2014). Only one other study to our knowledge has utilized nitrate sensors for numerical model evaluation (Jensen & Ford, 2019) - though their evaluation metric was concentration rather than hysteresis. Of the numerical models that utilize hysteresis as an evaluation metric, they are presently restricted to sediment transport (Mahoney et al., 2020), which typically only experiences mobilization, making discharge-concentration behavior more predictable. Here, we present an application of hysteresis modeling and

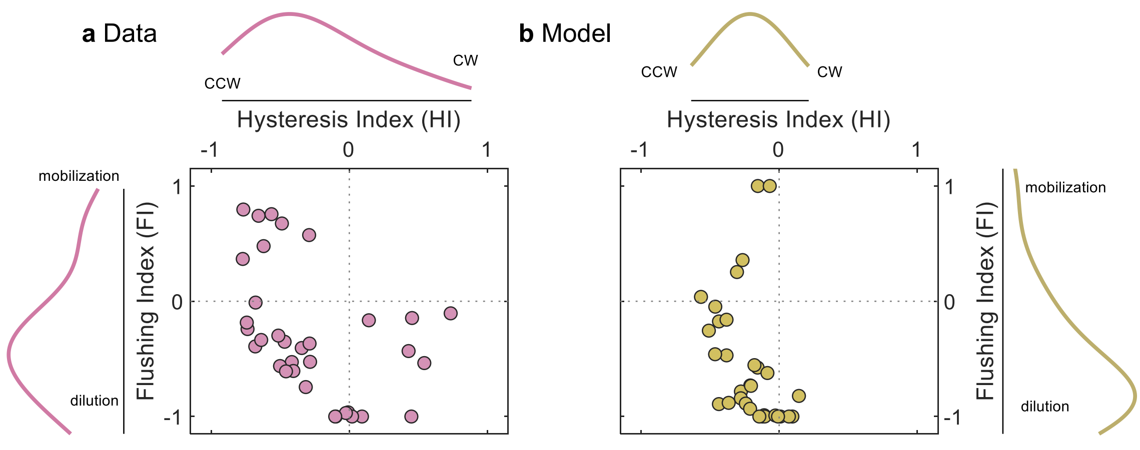

Figure 9. Nitrate flushing and hysteresis biplots for (a) daily averaged data results and (b) the best numerical model simulation. On average, the numerical model matches the general event flushing and hysteresis behavior observed in the data. CW = clockwise and CCW = counterclockwise.

in the next section we discuss how hysteresis inclusion improves the representation of hydrologic and nitrogen processes in our numerical model.

4.2.2. Hysteresis-Improved Uncertainty Estimation

The original model parameterization contains a substantial portion (87%) of high-frequency sensor data (Figure 10). However, uncertainty bounds are wide, and while the model primarily captures seasonal trends, predictions are less certain at the daily to-weekly scale. In this section, we integrated HI_N and FI_N parameters to the modeling objective criteria with the aim of reducing prediction uncertainty. After excluding parameter sets which failed to correctly model more than half of chemodynamic events (i.e., $|HI_N|$ and $|FI_N| > 0.10$), the hysteresis-improved model reduced the width of the original uncertainty bounds by 43 ± 12% while still encapsulating 71% of sensor data.

A series of events in February 2018 fell outside of the range of modeled predictions, disproportionately impacting statistics (Figure 10). This month had a series of large events in rapid succession (see Events 11 to 15 in Figure 3) as well as the largest flows and nitrate concentrations observed during the period. The high degree of aquifer wetness could have activated nitrate sources that are typically not linked to the spring (see our discussion on the importance of antecedent connectivity in Section 4.1.3). Potential sources include accumulated nitrate within soil zones that only contribute when wetted from an elevated water table or nitrate originating from combined sewer overflows in highly urbanized Lexington, KY (Figure 1). Both soil nitrate storage and sewer overflows are plausible sources given prior data and our understanding of the system. Regarding the first point, stable isotope data of Royal Spring nitrate ($\delta^{15}N_{NO_3}$ and $\delta^{18}O_{NO_3}$) from February 2018 averaged +6.12‰ and −0.11‰, respectively, which represents a nitrate source that is a combination of soil nitrate and manure and septic waste (Husic et al., 2020). To the second point, the City of Lexington is spending over half a billion dollars to repair failing wastewater infrastructure (UKCAFE, 2011), which has been identified as a primary detriment to water quality. Transient events like sewer overflows are not explicitly considered as inputs, as they are hard to reliably predict, thus our model may underestimate nitrate concentrations when such events occur.

Beyond reducing the width of nitrate uncertainty bounds, the hysteresis-improved model also constrained estimates of nutrient export from Royal Spring. The original model simulated quick-, intermediate-, and slow-flow nitrate loading fractions of 0.10 ± 0.04, 0.51 ± 0.24, and 0.39 ± 0.24, respectively, compared to hysteresis-improved fractions of 0.10 ± 0.02, 0.49 ± 0.11, and 0.41 ± 0.11, respectively. While the relative magnitude of the contributions is unchanged, the uncertainty (standard deviation) in the loading estimates is reduced by half, on average. This reduction in uncertainty can have implications for meeting potential loading targets at the spring. In this case, 90% of nitrate export occurs primarily via intermediate and slow pathways that represent the leaching of nitrate through the subsurface, which provides a clearer target for management. Further, the improved model also reduced parameter bounds (by 36 ± 20% on average; Table S2 in Supporting Information S1). The parameter space represents how the model interprets routing and storage of material, the principal drivers of hysteresis. The

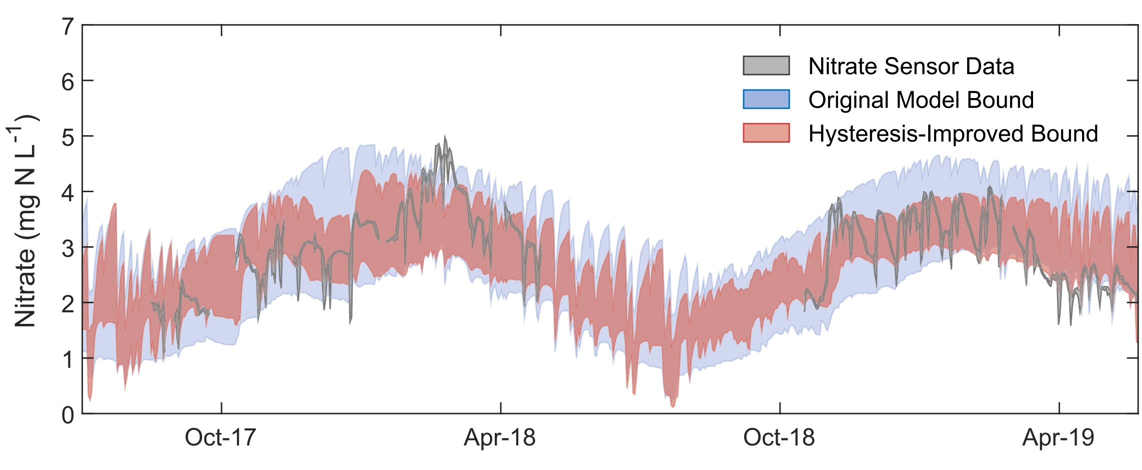

Figure 10. Numerical modeling results and high-frequency sensing data of nitrate at Royal Spring. The original model prediction bounds (blue shading; Husic, Fox, Adams, Ford, et al., 2019) are presented along with the hysteresis-improved prediction bounds (red shading; this study). The hysteresis-improved model represents a 43 ± 12% reduction uncertainty band width.

physical routing parameters with the greatest reduction in variability were X (fraction of rainfall as concentrated recharge), k_{soil} (soil percolation coefficient), and Q_{PUMP} (aquifer pumping rate). Importantly, the way in which the parameters changed after calibrating to nitrate hysteresis indicated that more rainfall was routed directly to concentrated recharge, the percolation of water through the soil and epikarst was faster, and that less groundwater is pumped from the aquifer. The sensitive storage and reactivity mechanisms included $V_{S,MIN}$ (soil field capacity), first-order rate constants (k_{DEN} and k_{NITR}), and the winter and summer nitrate recharge concentrations ($C_{NO_3^-(W)}$ and $C_{NO_3^-(SU)}$). These changes indicate more active biogeochemistry (i.e., larger first-order rate constants) and more nitrate accumulation in the reactive soil reservoir, which can later be connected to karst flow paths.

This work bridges the need for sensing water quality dynamics at the frequencies at which they occur (Rode et al., 2016) and numerically modeling processes which we cannot observe. Hysteresis indices provide an opportunity for this, and we show an example of their integration into numerical model evaluation. Integrating hysteresis into our model improved performance and process representation, and also reduced prediction and parameter uncertainty. While 10% of all original model realizations (n = 1,382 total realizations) accurately predicted the general behavior of both HI_N and FI_N, only ~1% correctly predicted the strength of more than half of the strongly chemodynamic events (i.e., $|HI_N|$ and $|FI_N| > 0.10$). By retaining fewer solutions, the posterior parameter space will shrink, thus leading to a reduction in equifinality—the condition whereby many different (sometimes contradictory) parameter sets yield satisfactory model simulations (Beven, 2006). A future area of improvement of the model structure is the time-scale at which it operates, which is presently limited to daily (Husic, Fox, Adams, Ford, et al., 2019). Thus, much of the dilution and subsequent concentration of the nitrate signal, which occurs at the sub-daily timescale, is not captured by the model. However, daily averaging the high-frequency data was still helpful to elucidate valuable insights to system function and to mitigate modeled uncertainty. We have shown that the integration of high-frequency data into numerical water quality models is a promising endeavor, particularly if integrated into models where the inputs, parameters, and structure are well recognized.

5. Conclusions

This work aimed to answer the question "what processes lead to observed nitrate behavior?" in karst settings. This question is often at the forefront of informing land management decisions. Our sensing and modeling results emphasize the importance of antecedent connectivity in linking nitrate-rich unsaturated storage zones, such as the soil and epikarst, to the spring. In this way, management of nitrate for karst springs will follow different strategies than used for surface systems. In the surface, baseflow remediation is done through in-stream restoration, which is not practical (or possible) for karst as conduits are located in complete darkness, are physically inaccessible, and often are not mapped with certainty. Peak flow mitigation in surface streams can be done by constructing detention basins, which again is not practical for karst systems as loading to the spring occurs primarily as leaching via fractures and matrix pores rather than by quickflow via sinkholes and swallets.

If water quality at a spring is the primary management goal, land managers should focus on mitigating subsurface nitrate leaching rather than slowing surface runoff. Ways to mitigate nitrate leaching include strategic application of fertilizer to minimize losses and planting cover crops to uptake existing nitrate within the soil, which we hypothesize as the primary nitrogen source to Royal Spring. However, nitrate legacies that have accumulated for decades in the soil and epikarst may continue to be exported for years after a land management change is implemented. Thus, while land managers can improve the timing of nutrient application, the benefits from that change may not be immediately apparent (Van Meter et al., 2018). Taken together, we have showed the potential of high-frequency sensing and numerical modeling to reduce uncertainty regarding dominant nitrogen processes and ultimately inform management to improve water quality.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Meteorological data, high-frequency sensing data, numerical modeling code, and modeling results will be openly accessible at the Open Science Framework (at: http://www.doi.org/10.17605/OSF.IO/XW4K9).

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